

1).

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problem1
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem1> node server
Server is listening on port number : 8000
```

← → ↻ ⓘ localhost:8000

NodeJS AJAX Example

Get Hello

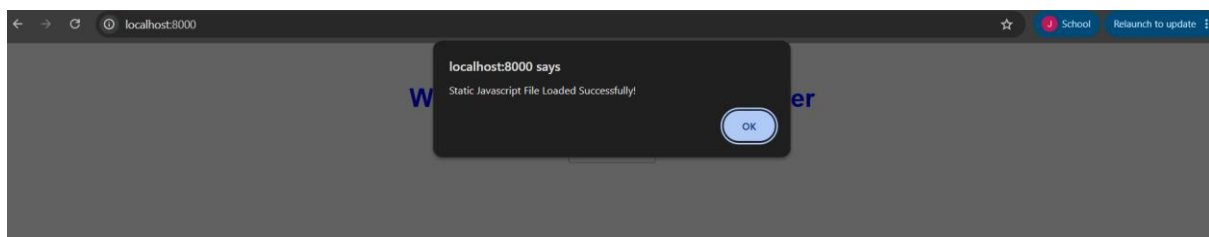
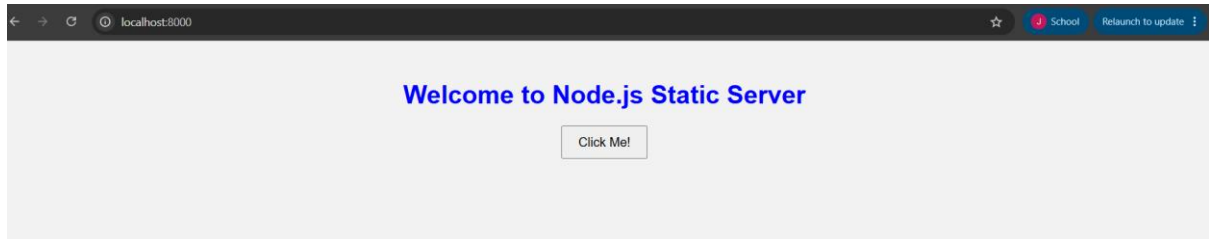
← → ↻ ⓘ localhost:8000/gethello

Hello NodeJS!!

2).

```
▼ problem2
  > node_modules
  ▼ public
    <> index.html
    JS script.js
    # style.css
    {} package-lock.json
    {} package.json
    JS server.js
```

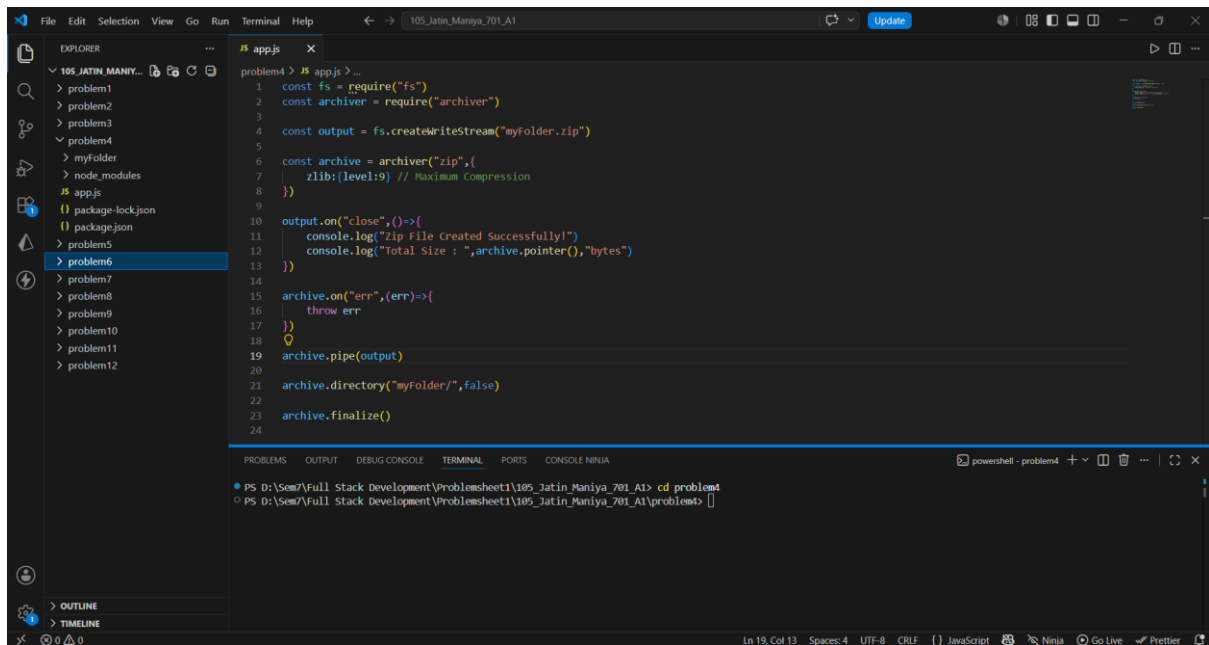
```
● PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problem2
○ PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem2> node server
Server is listening on port number 8000
```



3).

```
● PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem3> node app
===== College Chatbot =====
Type 'bye' to exit.
You : hi
Bot : Hello! Welcome to ABC College.
You : What courses are available?
Bot : We offer BCA, B.Sc IT, B.Com and MBA.
You : When are admissions open?
Bot : Admissions are open from June to August.
You : Where is the college located?
Bot : ABC College is located on College Road, Surat, Gujarat.
You : Is hostel available?
Bot : Yes, hostel facilities are available for both boys and girls.
You : thank you
Bot : Thank you. Have a nice day!
```

4).

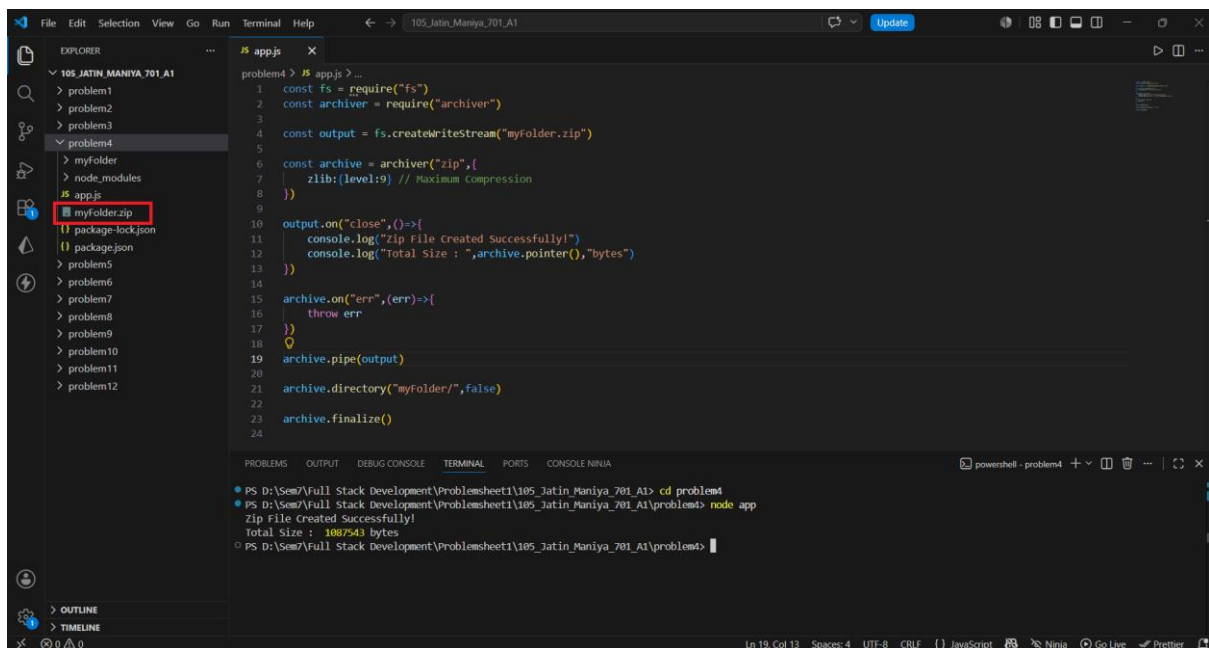


The screenshot shows the VS Code editor interface. The Explorer sidebar on the left displays a project structure with folders like '105_JATIN_MANIYA_701_A1', 'myfolder', and 'node_modules'. The file 'app.js' is selected. The main editor area shows the following JavaScript code:

```
1 const fs = require("fs")
2 const archiver = require("archiver")
3
4 const output = fs.createWriteStream("myfolder.zip")
5
6 const archive = archiver("zip",{
7   zlib:{level:9} // Maximum Compression
8 })
9
10 output.on("close",()=>{
11   console.log("Zip File Created Successfully!")
12   console.log("Total Size : ",archive.pointer(),"bytes")
13 })
14
15 archive.on("err",(err)=>{
16   throw err
17 })
18
19 archive.pipe(output)
20
21 archive.directory("myfolder/",false)
22
23 archive.finalize()
24
```

The terminal window at the bottom shows the command prompt with the following commands and output:

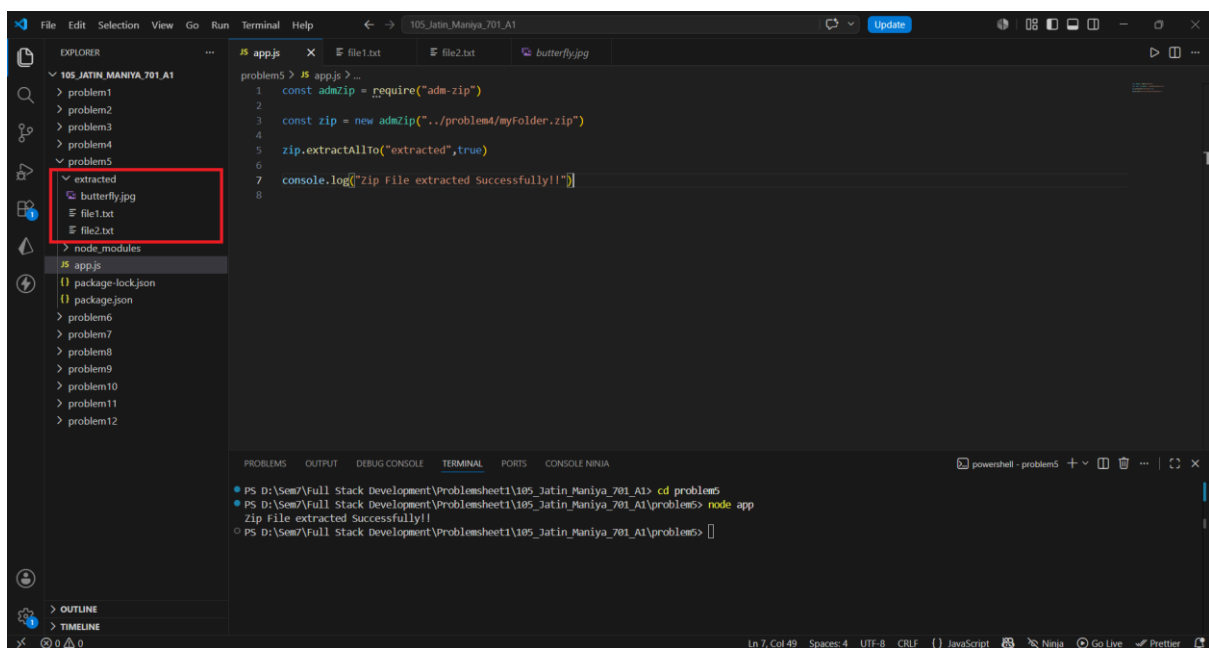
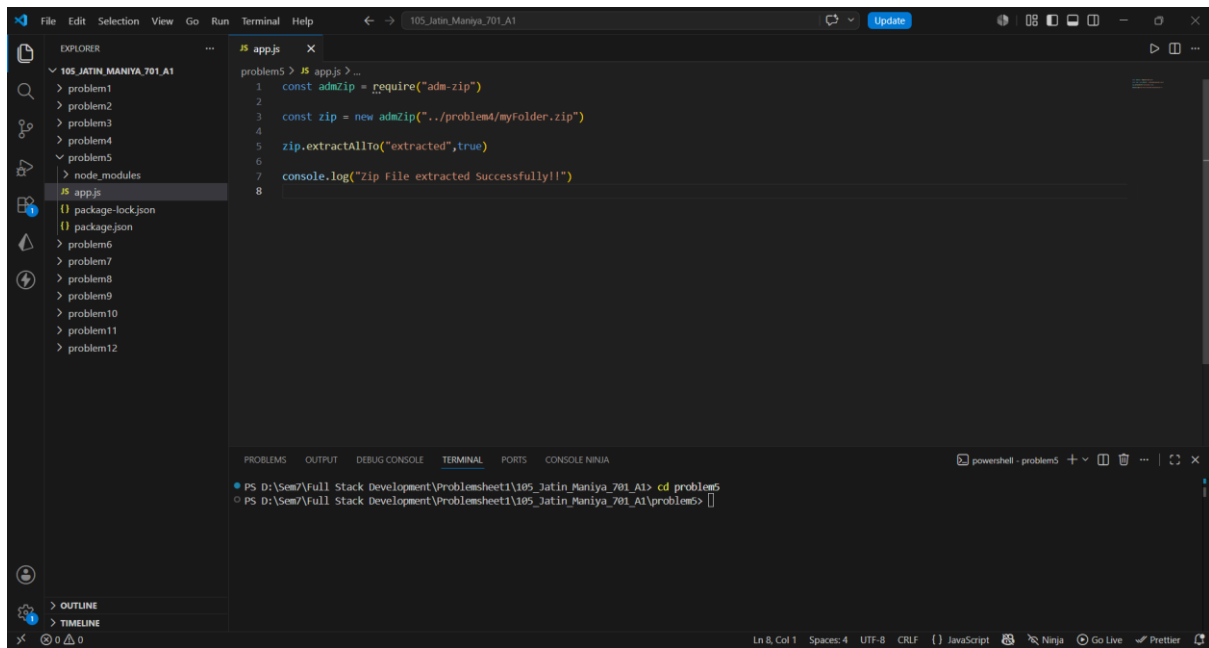
```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin Maniya 701 A1> cd problem4
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin Maniya 701 A1\problem4>
```



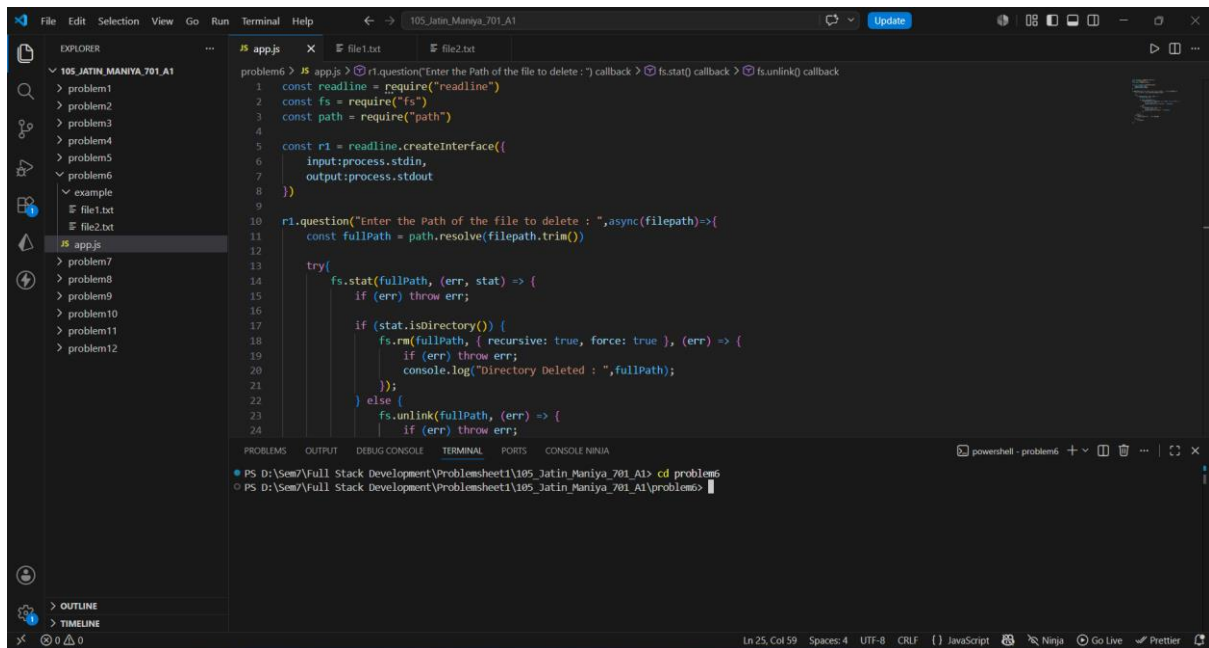
This screenshot shows the same VS Code editor interface after the code has been executed. The Explorer sidebar now includes the file 'myfolder.zip' under the 'myfolder' directory. The main editor area shows the same JavaScript code as in the previous screenshot. The terminal window now displays the output of the execution:

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin Maniya 701 A1> cd problem4
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin Maniya 701 A1\problem4> node app
Zip File Created Successfully!
Total Size : 1087543 bytes
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin Maniya 701 A1\problem4>
```

5).



6).

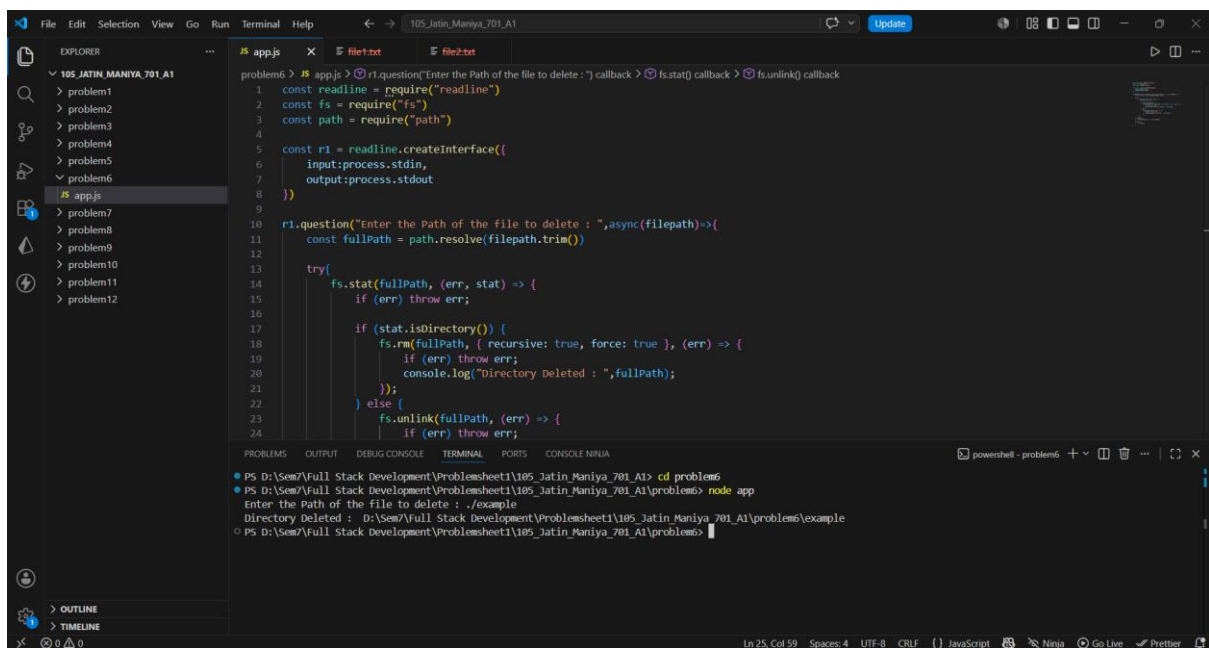


The image shows a VS Code editor window with the file explorer on the left, the editor in the center, and the terminal at the bottom. The file explorer shows a project named '105_JATIN_MANIYA_701_A1' with a folder 'example' containing 'file1.txt' and 'file2.txt'. The editor shows the 'app.js' file with the following code:

```
1 const readline = require("readline")
2 const fs = require("fs")
3 const path = require("path")
4
5 const rl = readline.createInterface({
6   input: process.stdin,
7   output: process.stdout
8 })
9
10 rl.question("Enter the Path of the file to delete : ", async (filepath) => {
11   const fullPath = path.resolve(filepath.trim())
12
13   try {
14     fs.stat(fullPath, (err, stat) => {
15       if (err) throw err;
16
17       if (stat.isDirectory()) {
18         fs.rm(fullPath, { recursive: true, force: true }, (err) => {
19           if (err) throw err;
20           console.log("Directory Deleted : ", fullPath);
21         });
22       } else {
23         fs.unlink(fullPath, (err) => {
24           if (err) throw err;
25         });
26       }
27     });
28   } catch (err) {
29     console.log("Error: ", err);
30   }
31 })
```

The terminal shows the following output:

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problems
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problems>
```



The image shows a VS Code editor window with the file explorer on the left, the editor in the center, and the terminal at the bottom. The file explorer shows a project named '105_JATIN_MANIYA_701_A1' with a folder 'example' containing 'file1.txt' and 'file2.txt'. The editor shows the 'app.js' file with the following code:

```
1 const readline = require("readline")
2 const fs = require("fs")
3 const path = require("path")
4
5 const rl = readline.createInterface({
6   input: process.stdin,
7   output: process.stdout
8 })
9
10 rl.question("Enter the Path of the file to delete : ", async (filepath) => {
11   const fullPath = path.resolve(filepath.trim())
12
13   try {
14     fs.stat(fullPath, (err, stat) => {
15       if (err) throw err;
16
17       if (stat.isDirectory()) {
18         fs.rm(fullPath, { recursive: true, force: true }, (err) => {
19           if (err) throw err;
20           console.log("Directory Deleted : ", fullPath);
21         });
22       } else {
23         fs.unlink(fullPath, (err) => {
24           if (err) throw err;
25         });
26       }
27     });
28   } catch (err) {
29     console.log("Error: ", err);
30   }
31 })
```

The terminal shows the following output:

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problems
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problems> node app
Enter the path of the file to delete : ./example
Directory Deleted : D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problems\example
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problems>
```

[illegible]

The screenshot shows the VS Code interface with the following components:

- File Explorer (Left):** Displays a project structure for '105_Jatin_Maniya_701_A1'. The 'package.json' file is selected.
- Code Editor (Center):** Shows the content of 'package.json':


```

{
  "name": "problems",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "start": "node index.js",
    "test": "echo \\\"Error: Running test cases...\\\" && exit 1",
    "dev": "nodemon index.js",
    "systemInfo": "node systemInfo.js",
    "delete-folder": "node delete.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
      
```
- Terminal (Bottom):** Shows the execution of 'npm run systemInfo' in the 'node - problem8' directory. The output includes:


```

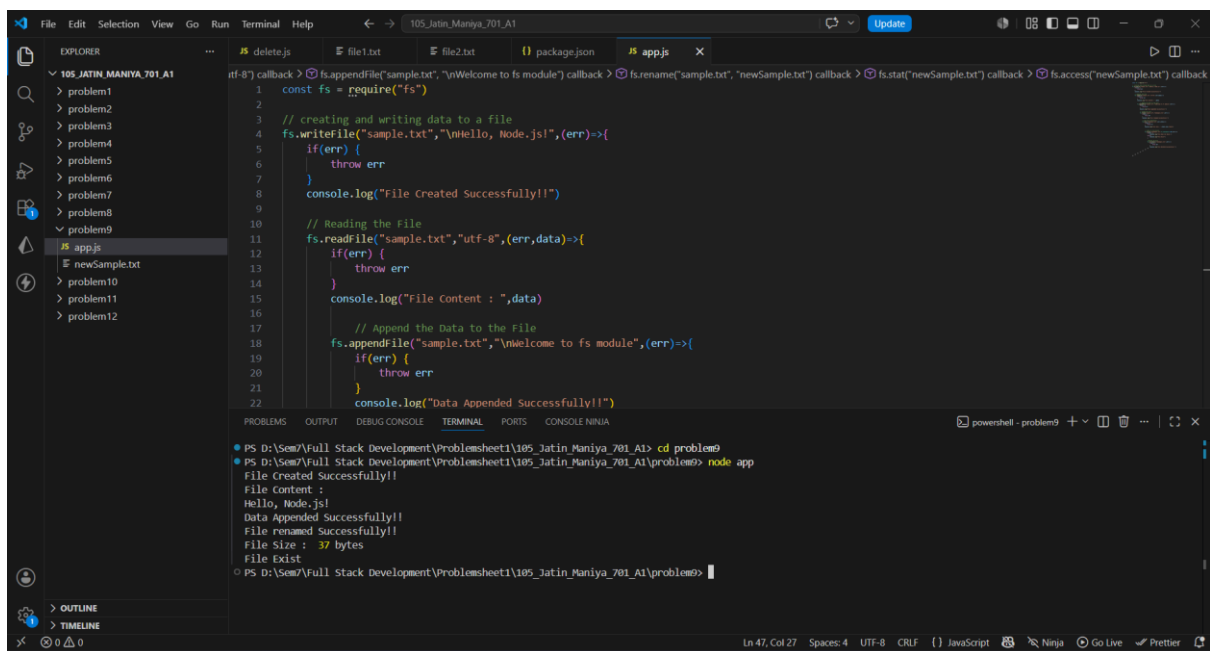
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem8> npm run systemInfo
> node systemInfo.js

Current Platform : win32
Current Time : 7/15/2026, 9:42:34 AM
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem8> npm run dev

> problem8@1.0.0 dev
> nodemon index.js

[nodemon] 3.1.14
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node index.js`
Server is listening on port number : 8000
      
```

9).

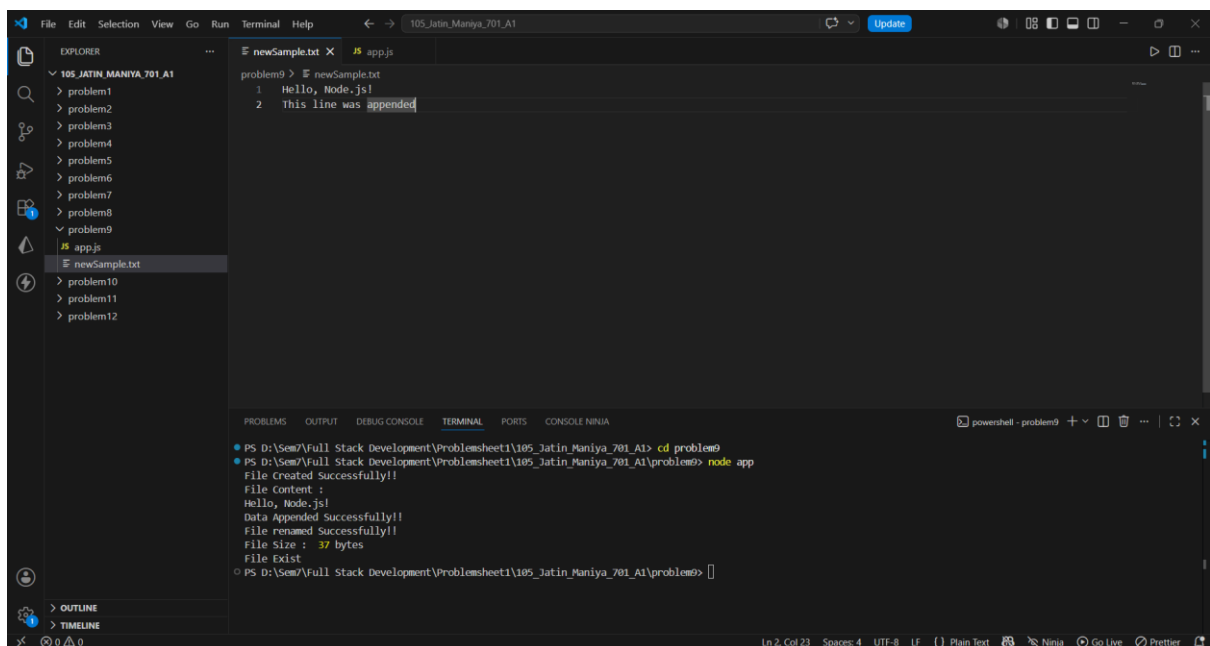


The screenshot shows the VS Code editor with the file explorer on the left displaying a project named '105_JATIN_MANIYA_701_A1'. The file 'app.js' is selected. The editor window shows the following JavaScript code:

```
1  const fs = require("fs")
2
3  // creating and writing data to a file
4  fs.writeFile("sample.txt", "Hello, Node.js!", (err) => {
5    if (err) {
6      throw err
7    }
8    console.log("File Created Successfully!!")
9  })
10
11 // Reading the File
12 fs.readFile("sample.txt", "utf-8", (err, data) => {
13   if (err) {
14     throw err
15   }
16   console.log("File Content : ", data)
17 }
18
19 // Append the Data to the File
20 fs.appendFile("sample.txt", "Hello, Node.js!", (err) => {
21   if (err) {
22     throw err
23   }
24   console.log("Data Appended Successfully!!")
25 })
```

The terminal at the bottom shows the following output:

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problem9
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem9> node app
File Created Successfully!!
File Content :
Hello, Node.js!
Data Appended Successfully!!
File renamed Successfully!!
File Size : 37 bytes
File Exist
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem9>
```



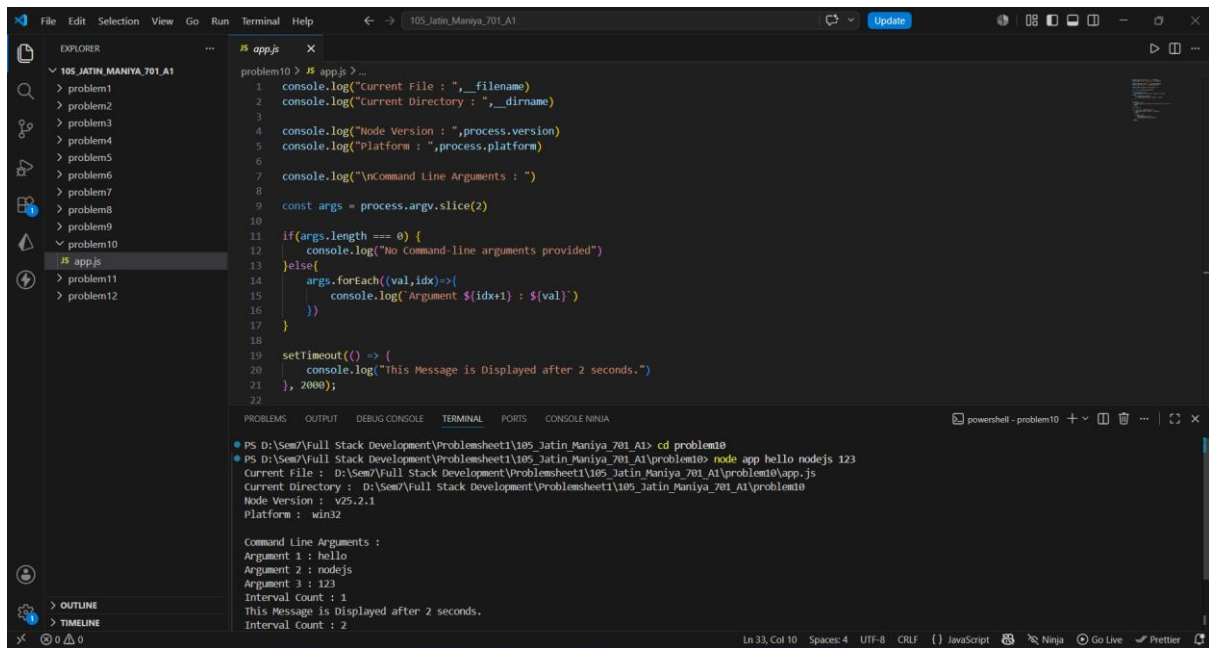
The screenshot shows the VS Code editor with the file explorer on the left displaying the same project. The file 'newSample.txt' is selected. The editor window shows the following text:

```
1 Hello, Node.js!
2 This line was appended
```

The terminal at the bottom shows the following output:

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problem9
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem9> node app
File Created Successfully!!
File Content :
Hello, Node.js!
Data Appended Successfully!!
File renamed Successfully!!
File Size : 37 bytes
File Exist
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem9>
```


10).



The screenshot shows the Visual Studio Code editor with a file named `app.js` open. The file contains the following JavaScript code:

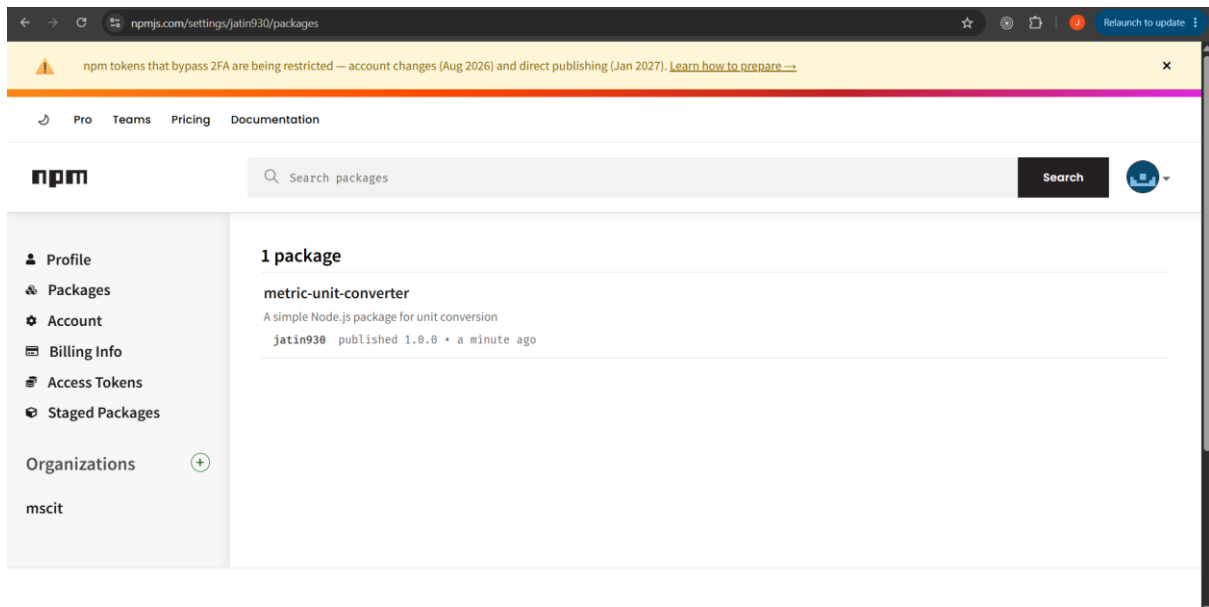
```
1 console.log("Current File : ", __filename)
2 console.log("Current Directory : ", __dirname)
3
4 console.log("Node Version : ", process.version)
5 console.log("Platform : ", process.platform)
6
7 console.log("\ncommand Line Arguments : ")
8
9 const args = process.argv.slice(2)
10
11 if(args.length === 0) {
12   console.log("No Command-line arguments provided")
13 } else {
14   args.forEach((val, idx) => {
15     console.log(`Argument ${idx+1} : ${val}`)
16   })
17 }
18
19 setTimeout(() => {
20   console.log("This Message is Displayed after 2 seconds.")
21 }, 2000);
22
```

The terminal output shows the execution of the script with the following details:

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problem10
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem10> node app hello nodejs 123
Current File : D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem10\app.js
Current Directory : D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem10
Node Version : v25.2.1
Platform : win32

Command Line Arguments :
Argument 1 : hello
Argument 2 : nodejs
Argument 3 : 123
Interval Count : 1
This Message is Displayed after 2 seconds.
Interval Count : 2
```

11).




```
1 const converter = require("metric-unit-converter")
2
3 console.log("10 km = ",converter.kmToMiles(10),"miles")
4 console.log("6.2 miles = ",converter.milesToKm(6.2),"km")
5
6 console.log("5 kg = ",converter.kgToPounds(5),"pounds")
7 console.log("11 pounds = ",converter.poundsToKg(11),"kg")
8
9 console.log("100 cm = ",converter.cmToInches(100),"inches")
10 console.log("39 inches = ",converter.inchesToCm(39),"cm")
11
12 console.log("25°C = ",converter.celsiusToFahrenheit(25),"°F")
13 console.log("77°F = ",converter.fahrenheitToCelsius(77),"°C")
14
```

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem11\metric-unit-converter> npm publish
Press ENTER to open in the browser...

+ metric-unit-converter@1.0.0
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem11\metric-unit-converter> npm i metric-unit-converter
added 1 package, and audited 2 packages in 3s

found 0 vulnerabilities
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem11\metric-unit-converter> node test
10 km = 6.21 miles
6.2 miles = 9.98 km
5 kg = 11.02 pounds
11 pounds = 4.99 kg
100 cm = 39.37 inches
39 inches = 99.06 cm
25°C = 77.00 °F
77°F = 25.00 °C
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem11\metric-unit-converter>
```

12).

```
1 function addtoCart() {
2   return new Promise((resolve)=>{
3     setTimeout(() => {
4       console.log("Product Added to Cart")
5       resolve()
6     }, 1000);
7   })
8 }
9
10 function makePayment() {
11   return new Promise((resolve)=>{
12     setTimeout(() => {
13       console.log("Payment Successful")
14       resolve()
15     }, 1000);
16   })
17 }
18
19 function placeOrder() {
20   return new Promise((resolve)=>{
21     setTimeout(() => {
22       console.log("Order Placed Successfully")
23       resolve()
24     }, 1000);
25   })
26 }
27
28 addtoCart()
```

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem12> node app
Product Added to Cart
Payment Successful
Order Placed Successfully
Shopping Completed
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem12>
```